

M1 IOT Evaluation

Suppose you downloaded a csv file named 'example.csv' which has the following description:

```
6,148,72,35,0,33.6,0.627,50,1
1,85,66,29,0,26.6,0.351,31,0
8,183,64,0,0,23.3,0.672,32,1
1,89,66,23,94,28.1,0.167,21,0
0,137,40,35,168,43.1,2.288,33,1
...
```

There are eight input variables and one output variable (the last column). You will build a model that learns to match rows of input variables (X) to an output variable (y). This is often summarized as $y = f(X)$.

Remark that the output variable is a Class variable (0 or 1).

1) Explain what does each line of the following code

```
dataset = loadtxt('example.csv', delimiter=',')
X = dataset[:,0:8]
y = dataset[:,8]
```

2) Using Keras, create a *Sequential model* which you name: model

The sequential model architecture has the following description :

- The model expects rows of data with 8 variables (the *input_shape=(8,)* argument).
- The first hidden layer has 12 nodes and uses the relu activation function.
- The second hidden layer has 8 nodes and uses the relu activation function.
- The output layer has one node and uses the sigmoid activation function.

3) Explain the following code :

```
model.compile(loss='binary_crossentropy', optimizer='adam',
metrics=['accuracy'])
```

4) Write the code that fit your model on your loaded data by calling the fit() function. We suppose that the training process will run for 200 epochs and that you consider a batch size of 10.

5) Give a clear definition of the number of epochs and the definition a batch size.

6) Write the code to evaluate the accuracy your model.